

UNIVERSITY OF THESSALY

POLYTECHNICAL SCHOOL OF

ELECTRICAL AND COMPUTER ENGINEERING

FINAL PROJECT OF MACHINE LEARNING

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1 Abstract

This project presents and analyzes data-driven predictive models for the energy use of appliances, as energy consumption forecasting contributes a lot in optimizing the resource planning and utilization in residential and commercial buildings. Forecasting was executed using ML and DL models, which specialize in handling time-dependencies and multivariate data management, along with a focus on leveraging data filtering, time series analysis and feature engineering techniques. The dataset, consisting of multivariate time series data over 4.5 months, was tailored to the needs of each model every time. Extend model review was implemented in various time-series models, such as traditional, machine and deep learning models, for selecting the top competitors for our analysis. Each of them has its pros and cons so the basic idea is the comparison of different approaches with the same goal: forecasting of buildings' energy consumption . The strongest competitor was the Transformer model given the fact of handling raw data without the need of feature engineering. In a surprising way, a machine learning model, GBM, was a top competitor, since there were more capable models for time series analysis, explaining an acceptable amount of variation in energy consumption.

2 Introduction

2.1 Problem Statement

Population growth, rapid industrialization, and the widespread adoption of energy-intensive technologies and lifestyles have increased energy consumption over the past decades. Among various sectors, buildings are responsible for a significant portion of global energy use, accounting for approximately 40% of total energy consumption. This includes residential, commercial, and industrial buildings, where heating, cooling, lighting, and appliance usage contribute to substantial energy demands. This growing demand creates challenges for efficient energy management, particularly in optimizing building operations and reducing costs.

Time series analysis is particularly relevant in this domain due to the sequential and cyclical nature of energy usage patterns. The nature of this analysis plays a crucial role in energy consumption forecasting by capturing patterns, trends, and seasonal variations in usage over time. These patterns, which often repeat over time, are important for accurately predicting future energy demand and ensuring efficient resource allocation. Forecasting energy consumption has become a powerful tool for estimating energy usage, specifically in buildings is essential to enhance energy efficiency, minimize waste, and support sustainable practices while ensuring occupants' comfort and reducing environmental impact.

This project, aims to evaluate the performance of different approaches in predicting energy consumption in buildings. By analyzing the performance of both Machine (ML) and Deep Learning (DL) models, the study explores the most reliable and precise approach for capturing temporal and complex patterns in energy usage.

2.2 Objectives

The core purpose of this project is to unroll the hidden dependencies that lie beyond the spectrum of numerical observations so we can effectively forecast household energy consumption, specifically on appliances energy usage, using machine and deep learning approaches. Analytically, the objectives are:

- Investigate the impact of multivariate features like house-metrics (temperature, humidity, etc.) and weather-metrics (atmospheric pressure, wind speed, etc.) on forecast accuracy.
- We aim to highlight the critical role of feature engineering in enhancing model performance, by comparing models trained on raw data with those utilizing engineered features.
- Providing insights for energy consumption patterns that are a product of the abstract concept of Human behavior → Device Operation → Energy consumption specifically for the appliances' energy usage.
- Compare and analyze the predictive power and efficiency of different ML and DL approaches to identify the most reliable model for building energy management applications.
- Explore the potential of Transformer models for energy forecasting due to their ability to handle long-range dependencies and complex multivariate relationships in sequential data.

2.3 Scope and Limitations

This study is designed to explore the energy consumption of a household based on data from a wireless sensor network and the weather data from a "nearby" airport's weather sensors combined over a period of 4.5 months, with observations recorded every 10 minutes. The scope of the project is defined by the following factors:

- **Data Nature:** The dataset includes energy consumption values alongside environmental features, such as indoor/outdoor temperatures and humidity levels, atmospheric pressure, wind speed, visibility, etc.
- **Models Applied:** The approaches that are examined are GBM for ML modeling, LSTM and GRU for sequential DL modeling, and, state-of-art architecture, namely Transformer.
- **Evaluation Metrics:** In order to compare the performance of each of the models, different performance evaluation indices are used here, like:
  1. Mean Squared Error (**MSE**)
  2. Root Mean Squared Error (**RMSE**)
  3. Mean Absolute Error (**MAE**)
  4. Coefficient of Determination (**R2**)

Regardless of the above, this project suffers from limiting factors which set boundaries in the general performance of the models. These restrictions are :

1. The length of continuous analyzed data. The provided dataset has a lifespan of only 4.5 months of appliances energy usage. That and only cripples potential research in this project for a long-term prediction of energy consumption. Being able to analyze -long lifespan- energy usage patterns, gives more insights in nested dependencies of appliances energy usage and features like the weather-related, season-related, day-related etc. Also, keeping in mind that the data observed from the sensors are captured in 10 minutes intervals, violates the assumption that the peak times may be wrong due to the unstability either of the sensors either from the fast overloads of the devices. !!!!!!!!!!————

2. An other limitation is the distant location of the airport's sensors, where the prediction of appliances energy usage could probably be better if the airport was closer to the house or even better if the house had equipment capable of weather observations.

3. The prediction of energy consumption that we are called to implement, is not based on every system unit that consumes energy, but only in appliances energy usage and lights. Light energy consumption ranges from 1 to 4% of the total since most lighting fixtures are LEDs and energy consumption of appliances represents between 70 and 79% of the monthly electrical energy consumption. Other factors like heating, cooling and ventilation, which include the remaining percentage of energy consumption and are directly depended on the sensor data, are not considered in dataset. Though, this project takes into account a big and acceptable portion of monthly energy usage.

4. High resource requirement because deep learning models, especially transformers, require significant computational power due to their large number of parameters and complex architectures. That matters why training these models on high-dimensional, multivariate time-series data demands GPUs or TPUs to achieve reasonable training times.

5. This research has not looked into the problem of optimal location of the wireless indoor sensors for improvement of the energy prediction. It is also possible that more, better and different types of sensors (e.g. occupancy detection) could help to improve energy prediction.

2.4 A Troubling Thought

The last and most important limitation is that the analysis was done for only ONE house. At first this does not seem that much of a problem, but it might be a solid reason that the energy forecast of the models cannot be generalized for any residence but only for this one. Occupancy and occupant's activity information could be useful to improve the prediction and find it's relationship with other parameters. Here comes the abstract concept of Human behavior $\rightarrow$ Device Operation $\rightarrow$ Energy consumption specifically for any appliance's energy usage.

Every household (detached house or building apartment) has some unique features like location, occupant's age, number of occupants, ownership of pets, building's geometry etc. When analyzing many houses we get a better range of "non-correlated" values because we keep in the radar the behavior of each house independently, therefore, the energy consumption of every different number of occupants behavior. The outcome of this behavior, how often and how much someone will use his available devices and the energy these device will consume, as a feature, will help the model to capture more dependencies in the actual features as the features themselves are a product of the human's behaviors. After the aforementioned, we summarize that each house has it's own unique set of data from smart sensors that captures the environmental metrics to human behaviors, that produce the energy consumption from their actions. The weather-related data captured from the house's and airport's sensors may fail to capture the true dependency between the energy use and the device operation in the home. Not having available information about the occupants' habits, we can't confirm what their future intentions may be, i.e. someone likes to keep the lights ON all day.

3 Model Review

This section evaluates a wide range of approaches for models that applied or could be applied in research forecasting energy consumption in residential buildings. Time series analysis plays a crucial role in these models because there are time-based dependances between values.

3.1 Traditional Time Series

Classical time series models, such as ARIMA (Autoregressive Integrated Moving Average, SARIMA(Seasonal ARIMA) and Exponential Smoothing (ETS, Holt-Winters) have been the cornerstone of time series forecasting for decades.

- **ARIMA** is a widely used statistical model for time series forecasting, incorporating autoregressive (AR), differencing (I), and moving average (MA) components. Works well with stationary data, interpretable, good for short-term forecasting but struggles with high-dimensional multivariate data
- **Seasonal ARIMA** (SARIMA) extends ARIMA by incorporating seasonality, making it more suitable for energy consumption data, which exhibits strong seasonal trends. Captures seasonality and trend components well but has high computational costs and requires extensive parameter tuning.
- **Exponential Smoothing** (ETS, Holt-Winters) uses weighted averages of past observations to predict future values. Effective for capturing trends and seasonality in time series data. They are simple and effective for small datasets but fail with complex interactions or non-linear behaviors.

In general, traditional models are interpretable and effective for simpler patterns but they lack of flexibility to model intricate dependencies or large datasets.

3.2 Machine Learning Models

Machine learning models such as Gradient Boosting Machines (GBM) and Random Forest (RF) are effective in predicting energy consumption by capturing complex, non-linear relationships between variables.

- **Gradient Boosting Machine** (GBM) is an ensemble learning technique that builds models sequentially, reducing prediction errors by combining weak learners. The paper found GBM to be highly effective in predicting appliance energy consumption. Handles non-linearity well, robust to overfitting with regularization but computationally expensive and requires hyperparameter tuning.
- **Random Forest** (RF) is an ensemble of decision trees that reduces variance by averaging multiple models. It has been widely used for energy forecasting due to its ability to handle high-dimensional data. Also, it has the advantage of handling missing data well and it is robust to overfitting. Those come with a price of usually being computationally expensive and less effective for extrapolation beyond training data.

3.3 Deep Learning Models

Deep learning models, such as LSTMs, GRUs, and CNNs, excel at capturing complex temporal and lasting dependencies in energy consumption data, making them well-suited for time series forecasting.

- **Long Short-Term Memory** (LSTMs) are specialized RNNs that excel in modeling long-term dependencies in sequential data. They have been widely used in energy forecasting by capturing short-term dependencies and handling missing data well. They excel in multivariate and long-range forecasts too, but can be prone to overfitting on small datasets. Generally, they require large datasets but can be computationally expensive.
- **Gated Recurrent Units** (GRUs) are simplified versions of LSTMs with fewer parameters, making them more efficient. Computationally efficient compared to LSTMs and perform well in time series forecasting. Main downside is the extensive hyperparameter tuning.
- **Convolutional Neural Networks** (CNNs), originally designed for image processing, have been adapted for time series forecasting by capturing local dependencies in data. Although, they are prone for long-range dependencies and usually require large datasets.

3.4 State-of-Art: Transformer Models

Transformers are deep learning architectures used in time series forecasting, NLP, and energy prediction models. Their core advantages come from self-attention, encoder-decoder architecture, and scalability.

1. **Self-attention :** It is a mechanism that allows transformers to focus on different parts of the input sequence when making predictions. Instead of just looking at the most recent data, like LSTM/GRU, self-attention enables global context awareness.

2. **Encoder-Decoder Framework :** The encoder converts an input sequence into a compressed context vector, that carries all the encoder's information, and the decoder generates an output sequence using this compressed information. For a deeper dive in Transformer's Components, the encoder stack has multiple layers of self-attention plus feedforward networks. The decoder stack uses masked self-attention plus encoder-decoder attention, where similarity is computed. The similarity between encoder and decoder states is computed using Scaled Dot-Product Attention. This dot-product similarity (QK^T) determines how strongly decoder queries align with encoder keys.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- Q = Queries from the **decoder**.
- K = Keys from the **encoder** (representing input sequence).
- V = Values from the **encoder** (context information).
- d_k = Scaling factor to stabilize gradients.

3. **Scalability** constitutes a big advantage of Transformers in comparison with other sequential models. Traditional sequential models (RNNs, LSTMs, GRUs) process data sequentially, meaning for each timestep, all the computations are applied before moving to the next. This limits parallelization and makes training slow. Transformers scale better because of their parallelization, long-sequence handling, and efficient attention mechanisms. Their scalability makes them ideal for large datasets and long time-series forecasting, like energy demand prediction.

Other transformers, such as the Temporal Fusion Transformers (TFT) and Informers, have shown superior performance in time series forecasting by leveraging self-attention mechanisms.

- The Temporal Fusion Transformer (TFT) what it does is : combines the sequence processing from LSTM or GRU and with the help of multi-head attention captures both temporal patterns and how multiple input features interact or influence each other simultaneously (e.g. high temperature and high humidity together might affect energy usage differently than high temperature but low humidity). Also, it stands out at forecasting multi steps ahead (e.g. hours or even week in the future). It uses a gating mechanism that is able to capture which features contribute the most at each time step. TFTs' biggest advantage is the intepretability they provide by the temporal importance they capture, showing which time steps are most influential in future predictions.TFT's encoder processes historical sequence with an LSTM/GRU network, capturing temporal depedencies in the past data and it incorporates an attention layer to highlight which time steps or features matter most. The decoder uses future-known covariates plus the hidden state from the encoder.

- Informers focus on long-horizon forecasting and with a reduced computation complexity. By using the ProSparse Self-Attention they are able to identify which are relatively insignificant so they can be pruned, thus. focusing only on the "active" queries and reducing the complexity.

4 Data Preprocessing

4.1 Data Description

In this project, the dataset is a combination of 2 different datasets collected from 2 different networks, one from the wireless sensor network from the house, and, the other from the airport station. Specifically:

- House-metrics: 9 temperature/humidity sensors are placed around the house covering every different room and one light controller.

- Weather-metrics: This dataset has measurements about the outside environment which are related to the state of the weather.

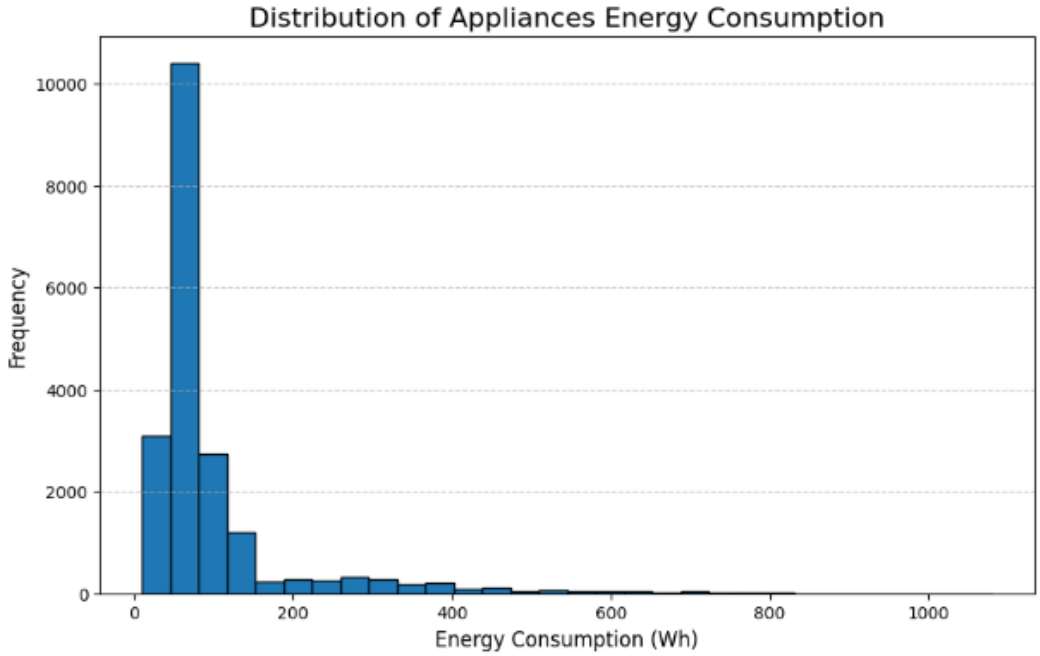
The dataset has a total of 19.735 rows and 29 columns. Those columns are Date(timestamp), Appliances(target), indoor metrics (T1 to T9 for temperature, RH1 to RH9 for relative humidity), weather metrics (Tout, Press_mm_hg, RHout, Windspeed, Visibility, Tdewpoint) and two random variables rv1, rv2. In addition, we have zero missing values.

The 10 min reporting interval was chosen to be able to capture quick changes in energy consumption. Also, the wireless sensor network's temperature and humidity recordings were averaged for the corresponding 10 min periods and merged with the energy data set by date and time. The time span of the data set is 137 days (4.5 months).

4.2 Exploratory Data analysis

EDA is a critical step in the data science field, because it provides insights into the dataset's structure, it's main characteristics, and the relationships between variables. Thus, this step is fundamental to acquire a first impression of the Appliances behavior based on the time, week, independent features etc and get a sense of what possible feature engineering techniques could be applied.

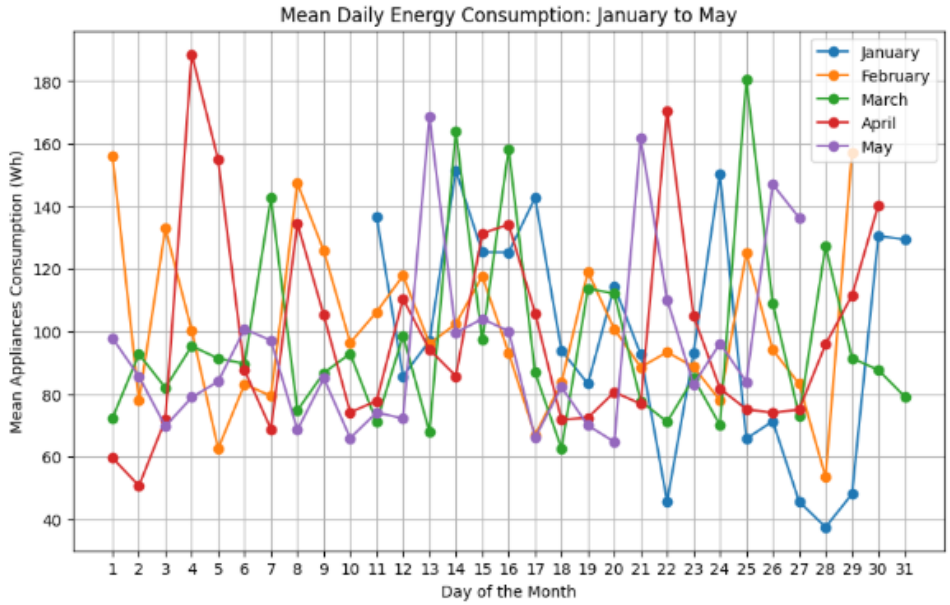
As a first look from pandas profiling, we got some interesting insights, like appliances distribution is skewed to the left and most frequently values range from 20Wh to 170Wh. This corresponds to the boxplot of appliances in which the median has a value of 60Wh. The lower whisker has a value of 10Wh, and the upper whisker has a value of 170Wh. It also shows that the data above the median is more dispersed and that there are several outliers. This shows that device energy consumption are around those values with some nonregular peaks. This is a concerning factor for energy prediction, because the nature of these peaks is possibly random. It can come from a sensor's malfunction, extreme sensor sensitivity (false observation), network instability or even some device overload at startup. So, the removal of all outliers is not the best choice because they might indicate valid, random, reasons for their existence. For ML tree-based models it is not a serious problem but for DL approaches it might add noise in pattern recognition.



Mean Daily Energy Consumption for all months

This plot shows the mean energy consumption of each day for all the months, so we can investigate how does the "Appliances" variability occur. As we see, all five months show big daily fluctuations (from as low as 60Wh up to 180 Wh), so there's no single month that's constantly higher or lower. Instead, each line jumps around quite a bit. This indicates that the monthly consumption doesn't differ much from month to month (may indicate constant 30d rolling avg). Also, we can't conclude

that there truly exists a clear seasonal pattern, because if we look at a single day of a month across all five months, we see some months are high while others are low.



- **Some Shared Peak Regions**
  We notice that many months have relatively high usage on or near the same days (e.g. days 2–3 or days 12–14 or around days 21–23). This might hint at recurring events (weekends or similar occupant routines), or perhaps random coincidence.

- **Likely Influencing Factors**
  While the charts only show raw daily usage, the spikes and dips could stem from seasonal weather (colder days could increase heating-related appliance use, or occupant behavior might shift). Occupancy/Behavior Changes (e.g., weekends or specific events could alter power usage). Random Events (unusual days with higher or lower appliance use).

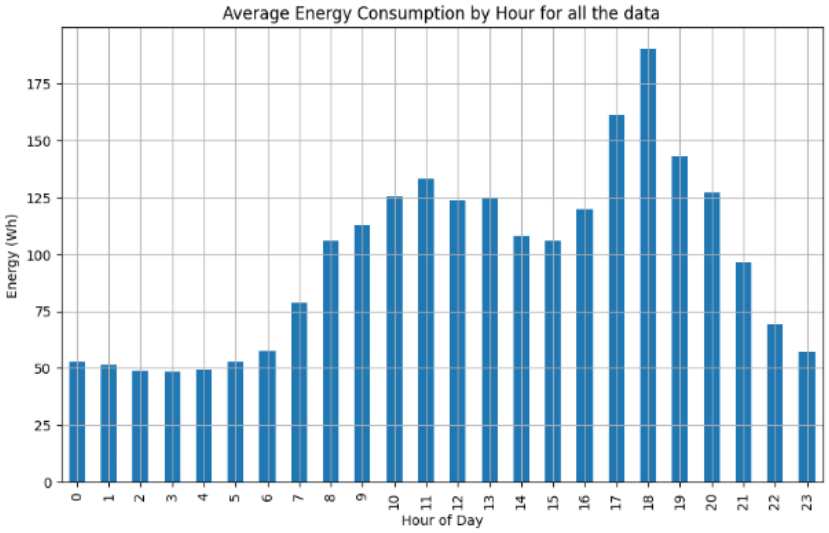
- **Hard to Pinpoint a Single Cause**
  Given the day-to-day variability, the spikes likely come from short-term changes-like heavier appliance use, cooking, entertainment, laundry, or weather shifts on specific days. At this point, it is crucial to tell that this dataset can possibly provide predictions only for short term trades.

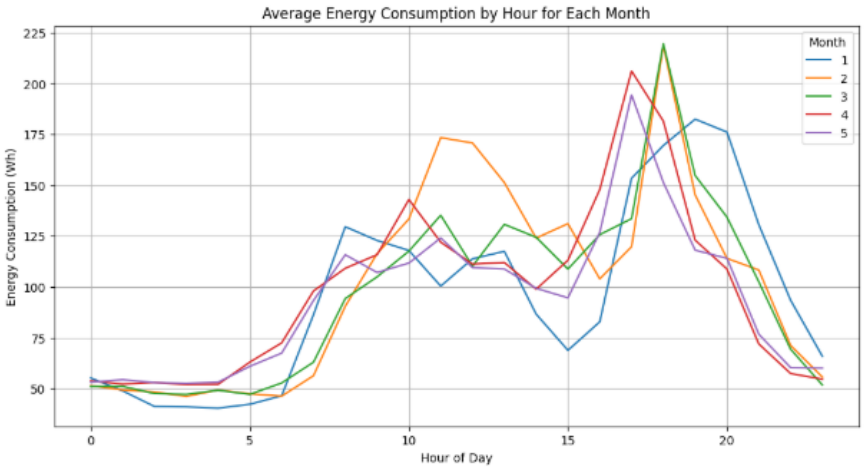
  For instance: The results indicate that almost all months have wide Daily fluctuations. Specifically, across all months the mean daily appliance consumption spans steeply at 60-180 Wh over the days. The "Appliances" indicate high day-to-day variability suggesting that occupant habits play a significant role in driving the daily changes.

  At the end of the day, even with every month behavior unrolled in front of us we could not end up in a summarized conclusion that will assist us much in data preprocessing or model's performance keeping always in mind that the appliances usage is mostly depended of occupants' habits.

Overall Daily Pattern (All Months Aggregated)

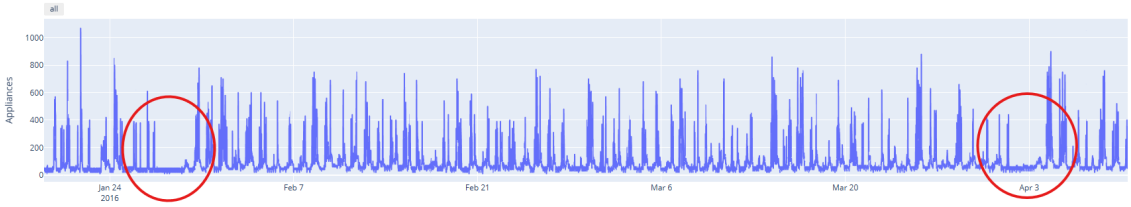
- Lowest usage overnight (0–5 AM). Consumption dips in the very early morning hours—often hovering around 45–60Wh between midnight and 5 AM. This suggests minimal appliance use when people are possibly asleep.
- Morning ramp-up (6–10 AM). Usage begins rising noticeably once people start their day—likely tied to breakfast cooking, showering, lights, and other morning routines.
- Mid-day plateau (10–15 PM). After the morning peak, there's often a plateau around 100–130Wh around late morning to early afternoon. This can reflect continued daytime occupancy, cooking, or moderate heating/cooling use.
- Largest spike in late afternoon/evening (16–19 PM). The biggest surge generally comes around dinnertime (17–19 PM), reaching 160–180Wh (or more). This is likely when people are home, cooking meals, and possibly running washers or other appliances.
- Gradual decline at night (20–23 PM). After the evening peak, consumption tapers off into the night as people finish evening activities and prepare for bed.





Interesting Insights

- We observe that at the end of January and early days of April, the Appliances usage was very limited. This could indicate that the occupancy level was low (maybe missing from the house).
- The correlation between the independent features is really high, due to the fact that the temperature will be much about the same especially when the sensors are included in the same floor or in the same region. This indicates the consequence of using HVAC.



4.3 Feature Engineering

Based on EDA's insights in this project, we used techniques in data transformation and time-series analysis techniques to preprocess raw data and extract meaningful features that enhance model performance. By applying normalization, and lag feature generation, we ensured that our time-series data was well-structured for training. Additionally, we leveraged statistical analysis, rolling window transformations, and decomposition methods to capture trends, seasonality, and anomalies. These preprocessing steps were crucial for improving the accuracy of our forecasting model, enabling it to learn temporal dependencies effectively. Analytically for data transformation and generation:

- **Normalization :** We applied Min-Max Scaling, which scales data to a fixed range because features have different ranges.
- **Lag Features:** Past values as additional features were added.
- **Rolling Window Features:** Moving averages and standard deviations were computed over a rolling time window.
- **Cyclical Encoding:** Used for handling cyclical/periodic features, such as hour of the day, day of the week, or month of the year

For the time series analysis, the techniques that will be mentioned below were applied but with no substantial result:

At first, time-series data decomposition was done into different components like trend, seasonality and residuals. This aimed in finding the overall direction of the series, repeating patterns and unexplained random variations. The results' uninterpretability lead us in DF-test. We applied the ADF statistic test which is used to check if a time series is stationary, meaning its statistical properties do not change

over time. The results showed that time series for Appliances is stationary, as indicated by the very negative ADF statistic, the p-value of 0.0, and the comparison of the test statistic with the critical values. This means that the series does not have a unit root, and its statistical properties like mean, variance, and autocorrelation are constant over time.

Correlation Matrix in Time-Series Analysis
A correlation matrix helps identify linear relationships between different features in a dataset. A correlation > 0.7 between two features, as a chosen threshold, leads to multicollinearity. Removing collinear features helps in improving model stability. In general, most sensors metrics (T1-T9, RH1-RH9) are highly correlated due to the fact that they are usually located on the same floor i.e., which are highly impacted by the HVAC usage at that time, as we mentioned before. Because of this, we keep the sensors that are strongly correlated with the target value and at the same time low correlated between each other.

Feature Importance A good strategy to keep the relevant features and reduce dimensionality of our dataset. Feature importance SHAP value is a powerful tool that shows us which features contributed most to the predictive target value. Also it is suited for our dataset because it handles non-linearity unlike simple correlation.

5 Results and Discussion

This section analyzes the forecasting performance of the applied models: GBM, LSTM, GRU, and time-series Transformers. Each of these models follows distinct methodologies, from the tree-based ensemble techniques of GBM to the sequential processing capabilities of deep learning approaches like LSTM and GRU, till the global awareness that TF can provide.

Feature engineering was tailored to each model to optimize their performance. EDA's insights guided us in selecting different feature engineering techniques like data transformation, feature importance, handcrafted lags and rolling values etc. in combination with the nature of the model's data handling. We keep a pattern of experimental tone in the phase of models' implementations, using different approaches on feature engineering and trying to catch as much as possible about the changes on model's behavior. Advanced details about these stages are shown in the notebooks.

The performance of the models in forecasting energy consumption was evaluated using three key metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R^2 Score. These metrics assess the accuracy of predictions and the models' ability to explain the variance in the data. The analytical progress is included in the respectively files.ipynb.

GBM Model

1st Trial: We consider all the features as "important" at first to see the model initial performance. Results were **MSE: 5595, RMSE: 74.46** and $\mathbf{R^2}$**: 0.4058** and that means our model captures some variance but leaves significant room for improvement.

2nd Trial: That improvement was accomplished with feature importance from SHAP. We kept the most significant features aligned with hyper-parameter tuning from GridSearch. Our greediness wanted more and the correlation matrix took place. Last attempt was with the diminished dataset from CM and the results were **MSE: 1449 , RMSE: 38.06 , R^2: 0.8248**.

Some extra trial: We tried implementing multi-step forecasting with GBM for the next 10min, 30min, 1hour. The model's performance was as shown :

- 10min Forecast - MSE: 5513.2 , RMSE: 74.25 , R^2: 0.4313
- 30min Forecast - MSE: 7571.73 , RMSE: 87.01 R^2: 0.2175
- 1hour Forecast - MSE: 7771.2 , RMSE: 88.15 R^2: 0.1938

```
final_gbm = GradientBoostingRegressor(
    learning_rate=0.01,
    max_depth=5,
    n_estimators=300,
    subsample=0.8,
    random_state=42
)
final_gbm.fit(X_train, y_train)
```

Figure 1: Best-hyperparameters for GBM based on GridSearch

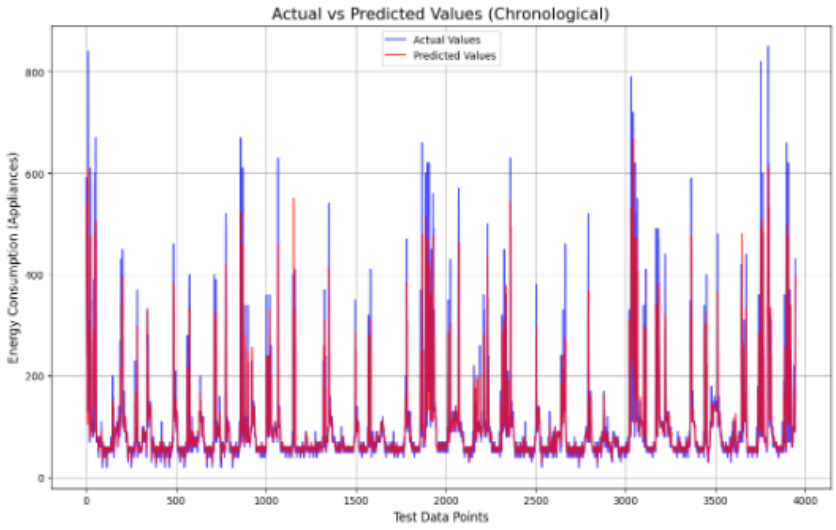


Figure 2: Best model predictions ($R^2 = 0.8248$)

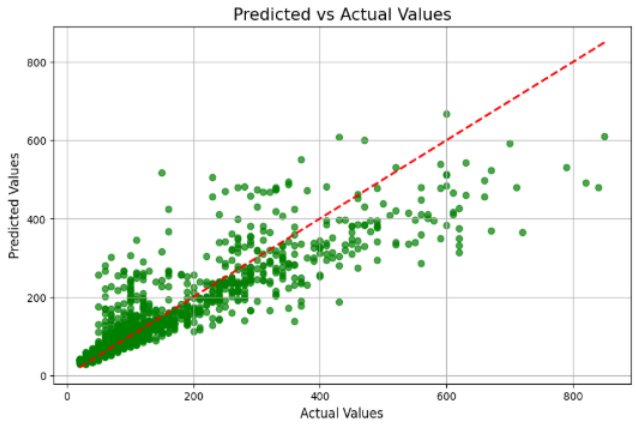


Figure 3: Residual plot of best GBM. The model seems to predict really good at range 60-400 with a bit underestimation of the actual values. After that range, the models seems it can't capture the patterns and maybe it is biased because of the rejection of the sensors information

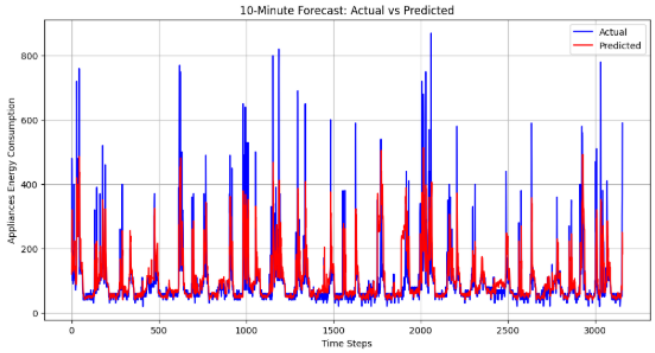


Figure 4: 10 min forecasting graph. It shows a good behavior considering the fact that it is a ML model that can't capture the behavior of the time existance.

DL Models (LSTM/GRU)

1st Trial: Keeping all the features plus adding some time-based features from feature engineering. The initial results were **RMSE: 57.70 , MSE: 3330 , MAE: 26.10 , R^2 Score: 0.5932** . We observe that we have much better initial variance explaination than GBM.

2nd Trial: Next, we applied LSTM again with much less features. We dropped the features based on Pearson's Corr Matrix. More details are shown in the respective notebook. The results were : **MAE: 2.46 , MSE: 21.78, RMSE : 4.67 R^2: 0.99**

3rd Trial: GRU implimentation based on the the same dataset as the first trial. Results : **MAE: 25.28, MSE: 3461.42, RMSE: 58.1 R^2: 0.577**. We observe that LSTMs(1) and GRUs are behaving almost the same. Also, for the **4rth trial** with less features same as in the 2nd trial, the results were : **MAE: 24.20 , MSE: 3291.9 RMSE: 57.37 , $R^2 = 0.5978$**. As we observe, the GRUs may have reached at the saturation point where there isn't much space for improvement.

5rth Trial: Autoencoders + GRU modeling with all the raw data features included. We used autoencoders because they excel at capturing temporal non-linear dependencies between the features. They act similarly with PCA but are

specialized in time series because they learn compressed latent representations of the time series. The results **RMSE: 63.03, MSE: 3972.63, MAE: 27.257, R^2 Score: 0.515** weren't as much optimistic as we expected.

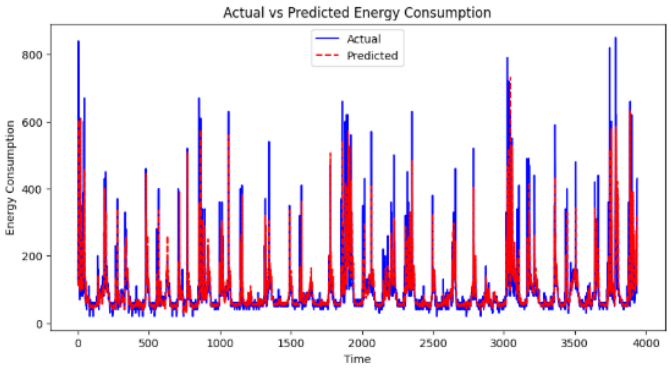


Figure 5: 1st trial with LSTM (R^2= 0.5932). Same for GRU(4th trial)

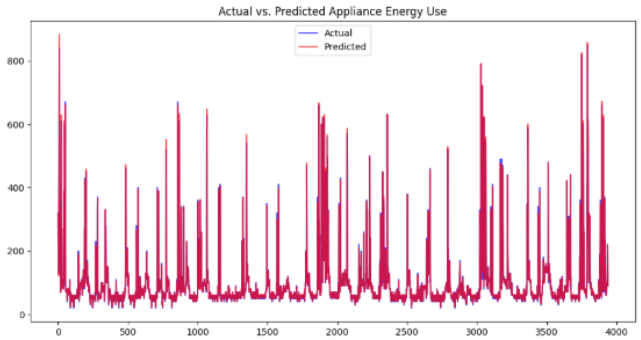


Figure 6: 2nd trial with LSTM. The predictions are almost aligned with the test values

Time Series Transformers

Transformers demonstrated outstanding performance in time series forecasting, achieving an impressive **R^2** value of **0.90** and **MAE: 0.1875 ,MSE: 0.0908, RMSE: 0.3014**. Remarkably, this was accomplished using raw data without extensive feature engineering, highlighting their ability to automatically extract complex temporal patterns. The power of multi-head attention further enhanced their capacity to focus on multiple aspects of the data simultaneously, enabling exceptional accuracy in energy consumption prediction task. Besides that, keeping in mind Transformers are computationally expensive and require extensive amount of training time, they are much faster than LSTMs and GRUs due to parallel data processing.

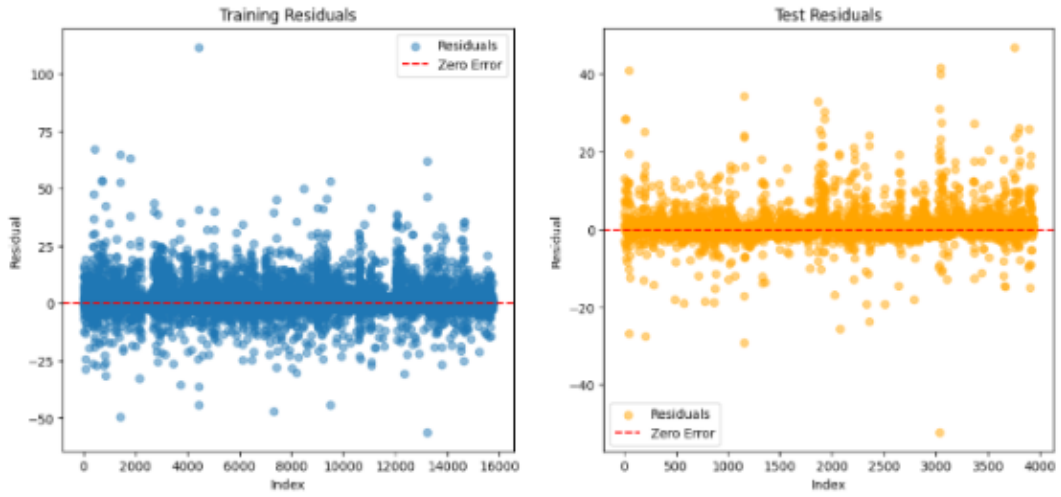


Figure 7: Residuals of the 2nd trial. The model with that huge variance explaination could indicate an overfitting in the test data. Although, the residuals are uniformaly distributed around the perfect fit, which means that there's no apparent trend in the errors which would indicate a severe systematic bias

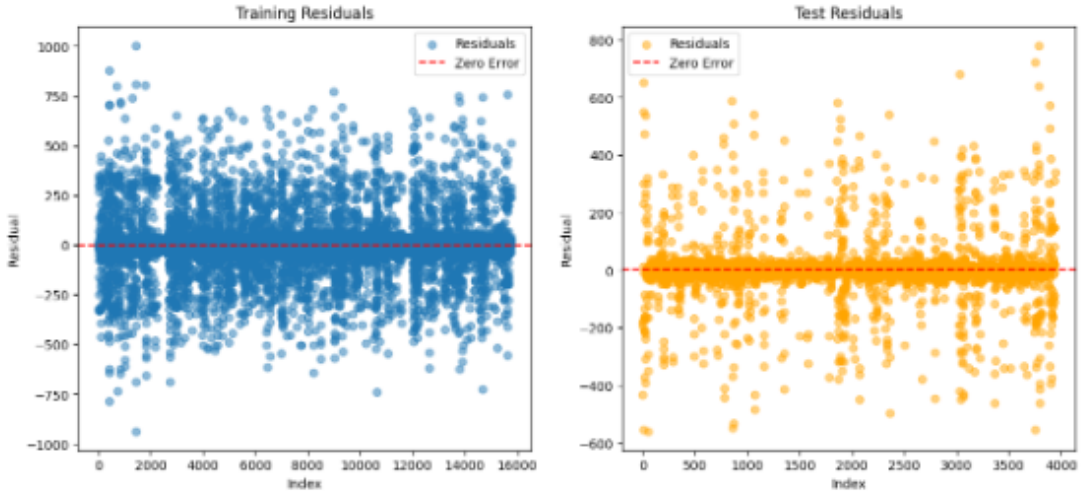


Figure 8: 3rd Trial, which is apparently the same both for LSTM's 1st trial, and, GRU. We see that there are outliers that reach higher or lower residual values. In those particular samples, the model's prediction error is larger than usual indicating an investigation of them because the model can't capture the true fit.

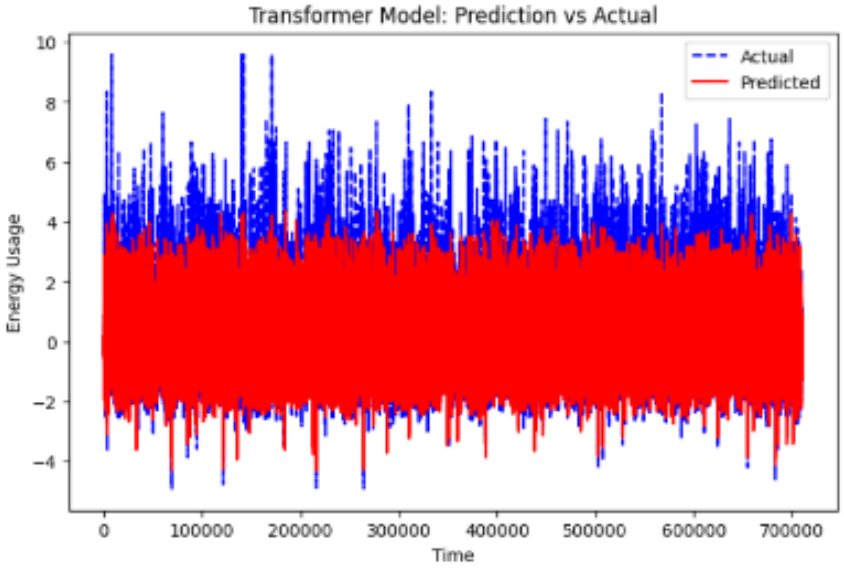


Figure 9: Transformer (R^2= 0.9)

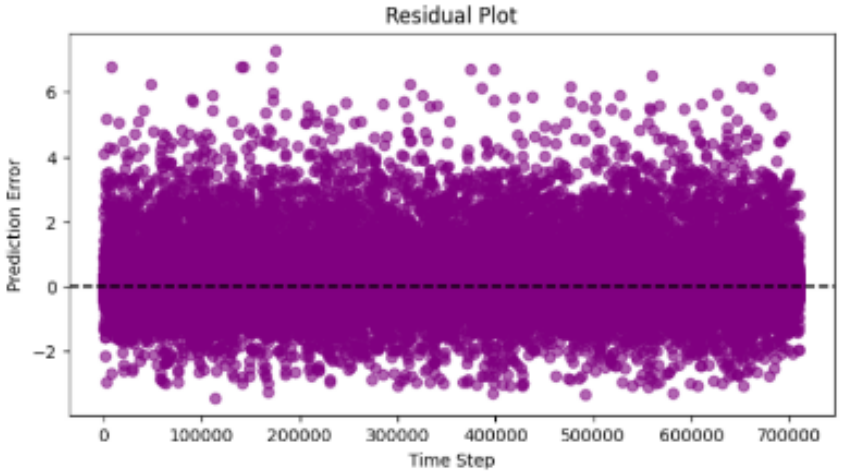


Figure 10: This residual plot shows that the TF model fails to predict some fluctuations in the energy demand e.g. sudden spikes or dips in energy consumption. Although, the residuals are distributed around this line, meaning the model does not have a strong bias

6 Conclusion

In conclusion, the GBM model had a really good performance on the first attempt with R^2=0.4058 on test data, which indicates an acceptable pattern recognition as an initial and baseline model. Although, with some feature engineering applications it had an extremely upgrade on the variance explaination reaching R^2=0.8248. From feature importance provided by SHAP, a really interesting observation was the fact that, the model captured mostly, the time-based generated features disusing with that way, the presence of the weather-related data from sensors. This extreme reliance on the time capturing, shows that the "Appliances" future usage from the devices is extremely dependent on how much the devices are already in operation. In simpler terms, the model's indicates that almost all sensors are useless for capturing the true prediction, and, that when a device is already operating, possibly it will continue consuming energy at the next time step. As for the LSTMs, they were able to fit perfectly the energy consumption for the current step, and, possibly they will perform just well for multi-step forecasting. Because the current paper's scope wasn't the LSTM's performance evaluation, this would be a future work for this specific research. From the other side, GRUs didn't show their significance in capturing the function that describes the "Appliances" true values. Also, they came at a saturation point that there were no significant changes by experimenting with this model. Besides that, by applying a state-of-art combination of Autoencoder and GRU, we didn't take better results. As for the transformers, they are indeed a state-of-art models because by feeding practically useless data captured from sensors that don't contribute to the description of the occupant's habits, they handled raw data significally good without the need of feature engineering, and accomplished to provide a description fuction that can apparently tell much about the "Appliances" energy consumption and thus, about the occupant's habbits. Finally, the GBM model is a really good choice for this specific task because of the interpretability and the easy implimentation it provides. As for the GRUs, beside the time consuming training they acquire, they are not as much interpretable and valuable in their results. For the transformer, it captured really good the real patterns between sensoring and Appliance using, but in a cost of non-interpretability.